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AI Resume Screening and Bias-Aware Ranking System

CSA6506 –Generative AI And Large Scale Models

Guided By:

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Abstract

- AI automates the resume screening process.
- It extracts skills, education, and work experience from resumes.
- The system matches resumes with the job description.
- Personal details are ignored to reduce bias in hiring.
- It ranks candidates fairly and helps recruiters make faster decisions.

Introduction

- Resume screening is the process of selecting suitable candidates for a job.
- Companies receive hundreds of resumes for a single job opening.
- Manual screening takes a lot of time and effort.
- AI automatically analyzes resumes and identifies qualified candidates.
- This system makes the hiring process faster, smarter, and fairer.

Problem Statement

- Manual resume screening is time-consuming.
- Recruiters need to review a large number of resumes.
- Human bias may affect candidate selection.
- Qualified candidates can be overlooked.
- An AI-based system is needed for faster and fairer recruitment.

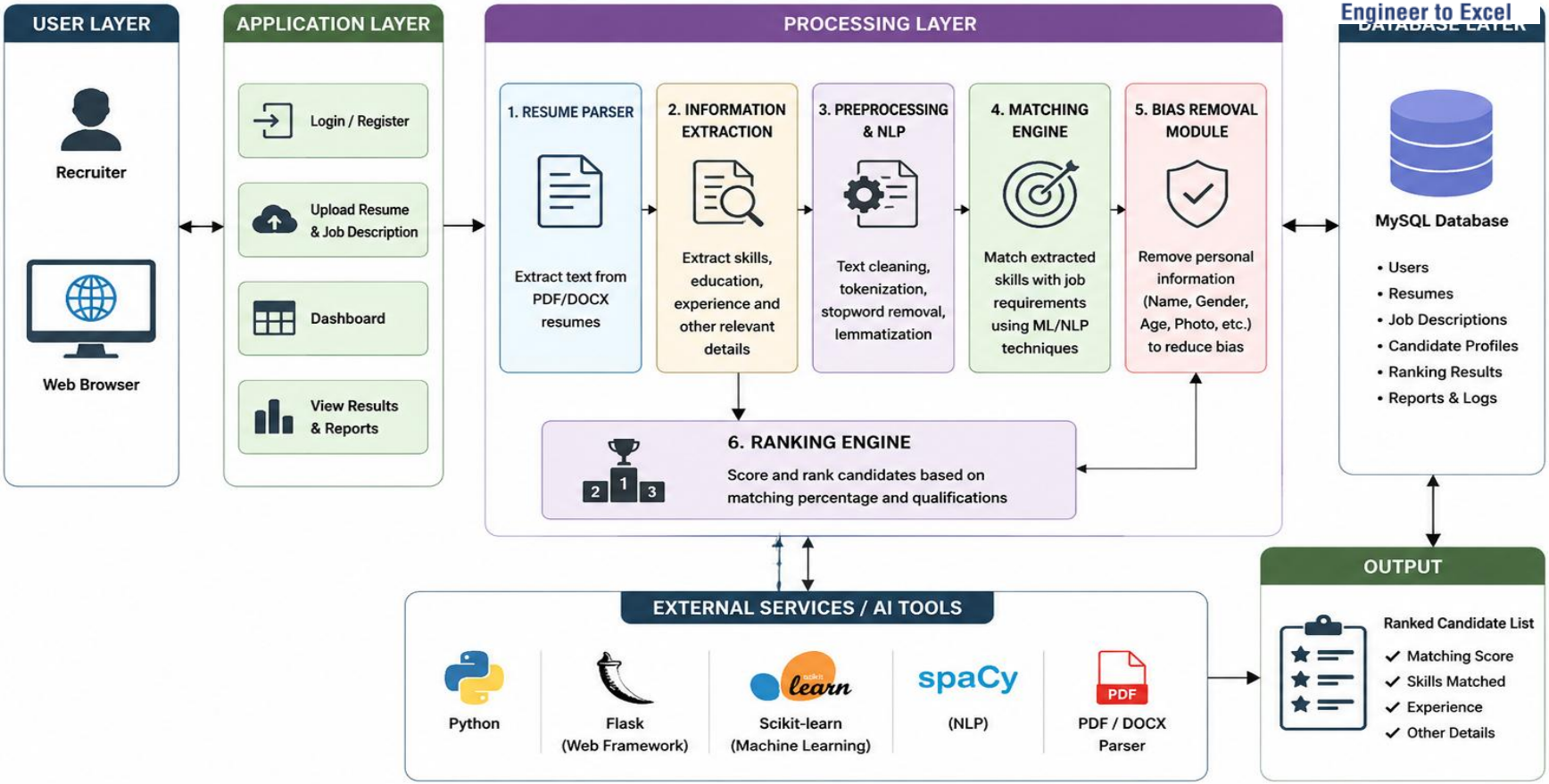
Objectives

- Automate the resume screening process.
- Extract skills, education, and work experience from resumes.
- Compare candidate profiles with job requirements.
- Reduce bias by ignoring personal information.
- Rank candidates based on their qualifications and suitability.

Literature Survey

S.No	Author & Year	Title of the Paper	Methodology	Key Findings
1	Wilson et al., 2024	Gender, Race, and Intersectional Bias in Resume Screening	Large Language Models (LLMs)	Identifies bias in AI resume screening and improves fairness.
2	Lo et al., 2025	AI Hiring with LLMs: A Context-Aware and Explainable Framework	Explainable AI (XAI) & Multi-Agent System	Provides transparent and accurate candidate evaluation.
3	Sabarirajan et al., 2025	AI-Based Resume Screening and Ranking Framework	Machine Learning & NLP	Automates resume screening and improves candidate ranking accuracy.
4	Pandit et al., 2026	Explainable Deep Learning for Intelligent Resume Screening	Deep Learning & Semantic Embeddings	Generates fair, explainable, and unbiased hiring recommendations.

System Architecture



Module 1: Resume Parsing and Candidate Profile Extraction

Engineer to Excel

- The system accepts resumes in PDF or DOCX format from candidates.
- AI extracts important information such as name, skills, education, experience, and certifications.
- The extracted data is converted into a structured candidate profile.
- The system organizes the information for easy processing and comparison.

Module 1-Output

Module 2: AI Resume Screening and Job Matching System

- The AI system analyzes the candidate's resume and the job description.
- It compares the candidate's skills, education, and experience with the job requirements.
- A job matching score is calculated based on the similarity between the resume and the job description.
- Candidates who meet the required qualifications are automatically shortlisted.

Module 2-Output

Module 3: Bias-Aware Ranking and Explainable Recommendation System

- The system removes personal details such as name, gender, age, photo, and address before evaluation.
- Candidates are ranked only based on their skills, qualifications, experience, and certifications.
- AI generates a fair ranking list without considering personal information.
- The system provides a clear explanation for each candidate's ranking and matching score.

Module 3-Output

Future Scope

- Integrate AI-based interview analysis to evaluate candidate communication and technical skills.
- Add LinkedIn and job portal integration for automatic profile and resume collection.
- Support multiple languages to analyze resumes from candidates across different regions.
- Develop a cloud-based recruitment platform for secure and remote access by recruiters.

Conclusion

- The AI Resume Screening and Bias-Aware Ranking System automates the recruitment process.
- It analyzes resumes and matches candidates with job requirements efficiently.
- The system reduces human bias by ignoring personal information during evaluation.
- It provides fair, accurate, and transparent candidate rankings.
- This solution helps recruiters save time, improve hiring quality, and make better recruitment decisions.

References

- Lo, F. P.-W., Qiu, J., Wang, Z., et al. (2025). *AI Hiring with LLMs: A Context-Aware and Explainable Multi-Agent Framework for Resume Screening.* CVPR Workshops.
- Sabarirajan, A., Chandramowleeswaran, G., Thamizharasi, D., et al. (2025). *Intelligent AI-Based Resume Screening and Ranking Framework for Unbiased and Scalable Recruitment Automation.* SCITEPRESS.
- Wilson, K., Sim, M., Gueorguieva, A.-M., & Caliskan, A. (2025). *No Thoughts Just AI: Biased LLM Recommendations Limit Human Agency in Resume Screening.* arXiv.
- Pandit, S. K., Chakraborty, K., Rajak, S., & Biswas, S. (2026). *An Explainable Deep Learning Framework for Intelligent Resume Screening Using Semantic Embeddings and Attention-Based Candidate Ranking.* Expert Systems with Applications.


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